

Complement Coupling in Time-Unfolded Gaussian Returns at a Quadratic Band-Merging Map:

Critical-Sibling Leakage, Floquet Replication, and a No-Go Theorem
for Naive Feshbach Identification

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Abstract

At the first algebraic band-merging parameter of $f_u(x) = 1 - ux^2$, a branch-isolated return of the exact Gaussian Markov operator gives a roots-of-unity ring whose principal radius accurately locates the observed small-noise bulk spectral edge. The missing step is a Feshbach theorem showing that the discarded complement only produces a controlled Schur self-energy. We prove that the most direct version of this step cannot work.

The first obstruction is dynamical. In the critical scaling $\delta/\sigma \rightarrow d$, the endpoint Gaussian pulls back to the conditioned Gaussian-quadratic profile $\Gamma_{d,t}$. Its exact reflection symmetry about the physical partition $b = u_c^{-1/2}$ implies

$$\frac{\|\mathbf{1}_{x>b}\mathcal{K}_\sigma^2 g_\sigma\|_{L^2(\text{critical})}}{\|\mathbf{1}_{x<b}\mathcal{K}_\sigma^2 g_\sigma\|_{L^2(\text{critical})}} \longrightarrow 1.$$

Thus the omitted critical sibling is not a Gaussian tail. The conclusion persists under the Riccati time weights because both branches occur at the same time slice. Splitting the final window into left and right pieces gives an exact operator identity $R^{\text{both}} = R^- + R^+$.

The second obstruction is algebraic. If $T = \mathcal{K}_\sigma^2$ and the full time-unfolded operator on $\mathcal{H}^k$ is $(\mathfrak{C}_k v)_j = T v_{j+1}$, then discrete Fourier transformation gives

$$\mathfrak{C}_k \simeq \bigoplus_{\ell=0}^{k-1} \omega_k^\ell T, \quad \det(\mathbf{I} - z\mathfrak{C}_k) = \det(\mathbf{I} - z^k T^k).$$

Hence every single physical eigenvalue of T generates a complete roots-of-unity ring in the time lift. After the canonical bipartite lift, one physical eigenvalue of $\mathcal{K}_\sigma$ generates $2k$ artificial rotations. A sector-free Feshbach completion of the local cyclic block can therefore recover only this replicated spectrum; it cannot by itself prove that the physical Markov operator has $2k$ distinct resonances.

We write the exact Schur self-energy for phase-dependent packet projections and show separately that global Perron/parity deflation does not commute with localization. Seven-noise sparse computations support the analytic obstruction. At $\sigma = 10^{-4}$, the right-sibling exit norm is 0.99738 of the retained left mass and its return eigenvalue is 0.99476 of the left return. Combining both branches changes the one-step radius from 0.75646 to 0.78982; the unrestricted endpoint return is 0.78992. These are ordinary floating-point diagnostics. The negative result does not invalidate the deterministic pole or the observed cloud. It replaces the naive scalar Feshbach target by a branch-resolved, Floquet-sector effective Hamiltonian in which peripheral extraction is built into the entrance and exit maps.

Keywords: Gaussian transfer operator; Feshbach map; Schur complement; Floquet decomposition; critical branch; spectral deflation; non-normal spectrum; quadratic map.

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1 Introduction

The small-noise bulk cloud at a quadratic band-merging map has acquired a substantial amount of structure. After the Perron and parity modes are removed, the deterministic two-step bulk determinant has an exact pole factor

$$\hat{D}_{0,\text{bulk},2}(z) = \frac{\mathcal{G}(z)}{1 - z^2/\lambda}, \quad \mathcal{G}(z) \neq 0 \quad (|z| < \lambda), \quad (1)$$

and finite Gaussian noise resolves that pole by an increasing resonance cloud [9, 10]. Endpoint row geometry supplies the intrinsic clock

$$N_\sigma = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad (2)$$

while the boundary word $CA(CB)^{k-1}$ supplies an exact length- k time ordering and a geometric cycle determinant [12, 14].

The preceding paper inserted the exact Gaussian dynamics into that time ordering [13]. Affine tangent channels possess a unique periodic Riccati packet, the final critical channel has an explicit conditioned Gaussian–quadratic limit, and time-labeled restrictions form a compact block-cyclic auxiliary operator. Its principal return generates an exact root ring. With $k = N_\sigma + 1$, the branch-isolated radius differs from the archived full bulk edge by only 5.64×10^{-4} at $\sigma = 10^{-4}$.

That result deliberately stopped before a Feshbach identification. If P_σ denotes the packet block and $Q_\sigma = \mathbf{I} - P_\sigma$, the desired effective operator contains the Schur self-energy

$$\Sigma_\sigma(\zeta) = P_\sigma \mathfrak{C} Q_\sigma (\zeta Q_\sigma - Q_\sigma \mathfrak{C} Q_\sigma)^{-1} Q_\sigma \mathfrak{C} P_\sigma. \quad (3)$$

A tempting strategy is to enlarge the packet windows slowly, use Gaussian tails to make the two off-diagonal couplings small, and then compare determinants on contours separated by the $1/k$ root spacing. This paper tests that strategy before attempting the difficult complement resolvent.

It fails for two independent reasons. First, the complement contains a second deterministic critical preimage with the same asymptotic mass as the retained branch. It is not part of the far Gaussian tail. Second, the full time-unfolded operator itself has exact roots-of-unity symmetry by discrete Fourier replication. Feshbach completion in the unfolded space recovers that replicated spectrum, not a count of distinct eigenvalues of the physical Markov operator.

These are useful negative results. They isolate exactly which statement is false without changing the analytic determinant pole, the endpoint rank, the deterministic cycle identities, or the numerical cloud. They also specify a corrected target: a two-branch effective Hamiltonian with an explicit Floquet phase, and with Perron/parity extraction incorporated before localization.

Main results and logical status

- C1. Critical-sibling theorem.** On symmetric $O(\sqrt{\sigma})$ windows about $b = u_c^{-1/2}$, the left and right pullback masses have ratio one. This follows analytically from the RH-18 limiting profile. It rules out an $o(1)$ off-block coupling estimate based only on Gaussian tails.
- C2. Exact branch splitting.** The return through a branch-complete final window is the operator sum of the left and right returns. Positivity orders the one-branch, branch-complete, and unrestricted finite-horizon returns.
- C3. Floquet replication theorem.** The full cyclic time lift is unitarily equivalent to k rotated copies of the physical two-step operator. Its determinant and spectrum therefore

contain exact root rings even if the physical operator has only one eigenvalue. Root counting in the lift is not physical resonance counting.

C4. Exact Feshbach map and no-go corollary. We write the phase-dependent Schur complement of the time lift. Even a successful small-self-energy theorem for this sector-free lift would only identify the replicated Floquet spectrum. A physical conclusion requires a sector condition or an equivalent spectral-parameter boundary map.

C5. Deflation–localization noncommutation. Removing the Perron/parity spectral projectors before localization changes every packet channel by an explicit rank-two term. Consequently those modes cannot be “removed after” a positive branch return without a commutator estimate.

C6. Floating-point audit. At the two cleanest tail points, $\sigma = 10^{-3}$ and 10^{-4} , the sibling exit ratios are 0.99030 and 0.99738. The right/left return-eigenvalue ratios are 1.00564 and 0.99476. The unrestricted endpoint return is already reproduced by the two-branch return to relative errors 4.21×10^{-4} and 1.92×10^{-3} . No interval or pseudospectral enclosure is claimed.

The corrected logical diagram is

$$\boxed{\begin{array}{l} \text{single branch local ring} \not\Rightarrow \text{small complement,} \\ \text{sector-free time-lift Feshbach theorem} \not\Rightarrow \text{physical cloud multiplicity,} \\ \text{two branches} + \text{peripheral extraction} + \text{Floquet sector,} \\ \Rightarrow? \text{ physical effective determinant.} \end{array}} \quad (4)$$

The first two non-implications are proved here. The final arrow is the next operator problem.

2 Map, kernel, and boundary scaling

Let u_c be the root in $(1, 2)$ of

$$u^3 - 2u^2 + 2u - 2 = 0, \quad (5)$$

and put

$$f(x) = 1 - u_c x^2, \quad r = u_c - 1, \quad \lambda = 2u_c r. \quad (6)$$

Numerically,

$$u_c = 1.543689012692076 \dots, \quad \lambda = 1.678573510428322 \dots \quad (7)$$

Write $S = f^2$ and

$$h(x) = \sqrt{\frac{1 - \sqrt{(1-x)/u_c}}{u_c}}, \quad b = h(1) = u_c^{-1/2}. \quad (8)$$

The distinguished boundary point $p_k = p_{2k}$ has clearance

$$\delta_k = 1 - p_k = C_b \lambda^{-2k} (1 + o(1)), \quad C_b > 0, \quad (9)$$

and the final source point is

$$s_k = h(p_k) < b, \quad b - s_k = \frac{\sqrt{\delta_k}}{2u_c} + O(\delta_k). \quad (10)$$

For $\sigma > 0$, the row-normalized folded Gaussian kernel on $[0, 1]$ is

$$P_\sigma(x, y) = \frac{\phi_\sigma(y - f(x)) + \phi_\sigma(y + f(x))}{Z_\sigma(x)}, \quad (11)$$

$$Z_\sigma(x) = \int_0^1 \{\phi_\sigma(y - f(x)) + \phi_\sigma(y + f(x))\} dy, \quad (12)$$

where

$$\phi_\sigma(t) = \frac{1}{\sqrt{2\pi}\sigma} e^{-t^2/(2\sigma^2)}. \quad (13)$$

The backward Markov operator is

$$(\mathcal{K}_\sigma g)(x) = \int_0^1 P_\sigma(x, y) g(y) dy. \quad (14)$$

Its kernel is smooth and strictly positive at fixed noise. We write

$$\mathcal{T}_\sigma = \mathcal{K}_\sigma^2 \quad (15)$$

for the two-step component operator. On $L^2(0, 1)$, $\mathcal{K}_\sigma$ is Hilbert–Schmidt and $\mathcal{T}_\sigma$ is trace class, so the Fredholm determinant identities used below apply.

At the intrinsic clock (2), the endpoint packet has width $O(\sigma)$, whereas its final pullback has width $O(\sqrt{\sigma})$. Let

$$p_\sigma = 1 - \sigma d_\sigma, \quad d_\sigma \rightarrow d > 0, \quad s_\sigma = h(p_\sigma), \quad (16)$$

and define the endpoint observable

$$g_\sigma(y) = \exp\left[-\frac{(y - p_\sigma)^2}{2\sigma^2 t_\sigma^2}\right], \quad t_\sigma \rightarrow t > 0. \quad (17)$$

The RH-18 critical-profile theorem states, locally uniformly in q , that

$$(\mathcal{T}_\sigma g_\sigma)(s_\sigma + \sqrt{\sigma}q) \rightarrow \Gamma_{d,t}(q), \quad (18)$$

where

$$A_d(q) = (\sqrt{d} - 2u_c q)^2, \quad B_d(q) = d - A_d(q), \quad (19)$$

$$\Gamma_{d,t}(q) = \frac{t}{\sqrt{1+t^2}} e^{-B_d(q)^2/[2(1+t^2)]} \frac{\Phi\left[\frac{\sqrt{1+t^2}}{t} \left(A_d(q) + \frac{B_d(q)}{1+t^2}\right)\right]}{\Phi(A_d(q))}. \quad (20)$$

The symmetry hidden in this formula is the first complement obstruction.

3 The critical sibling is an order-one channel

Put

$$q_b = \frac{\sqrt{d}}{2u_c}. \quad (21)$$

Then $A_d(q_b + s) = A_d(q_b - s)$ and therefore

$$\Gamma_{d,t}(q_b + s) = \Gamma_{d,t}(q_b - s). \quad (22)$$

Moreover, (10) gives

$$\frac{b - s_\sigma}{\sqrt{\sigma}} \rightarrow q_b. \quad (23)$$

For fixed $A > 0$, define symmetric physical windows

$$J_{\sigma,A}^- = \{s_\sigma + \sqrt{\sigma}q : q_b - A < q < q_b\}, \quad (24)$$

$$J_{\sigma,A}^+ = \{s_\sigma + \sqrt{\sigma}q : q_b < q < q_b + A\}. \quad (25)$$

Up to an $o(\sqrt{\sigma})$ displacement, these are the two sides of the physical partition b .

Theorem 3.1 (Equal critical-sibling mass). *For every fixed $A > 0$,*

$$\sigma^{-1/2} \|\mathbf{1}_{J_{\sigma,A}^-} \mathcal{T}_\sigma g_\sigma\|_2^2 \longrightarrow \int_{q_b-A}^{q_b} \Gamma_{d,t}(q)^2 dq, \quad (26)$$

$$\sigma^{-1/2} \|\mathbf{1}_{J_{\sigma,A}^+} \mathcal{T}_\sigma g_\sigma\|_2^2 \longrightarrow \int_{q_b}^{q_b+A} \Gamma_{d,t}(q)^2 dq. \quad (27)$$

The two limiting integrals are equal and positive. Consequently

$$\frac{\|\mathbf{1}_{J_{\sigma,A}^+} \mathcal{T}_\sigma g_\sigma\|_2}{\|\mathbf{1}_{J_{\sigma,A}^-} \mathcal{T}_\sigma g_\sigma\|_2} \longrightarrow 1. \quad (28)$$

Proof. Use $x = s_\sigma + \sqrt{\sigma}q$ in each squared norm. The Jacobian is $dx = \sqrt{\sigma} dq$, and the integration intervals are fixed in q . Locally uniform convergence in (18) therefore gives (26)–(27). Reflection $q \mapsto 2q_b - q$ and (22) identify the two integrals. Strict positivity of $\Gamma_{d,t}$ makes them nonzero. $\square$

Corollary 3.2 (No Gaussian-tail estimate for branch isolation). *Let $P_{\sigma,A}^-$ and $P_{\sigma,A}^+$ be multiplication by the two windows in (24)–(25). If the retained packet space contains $P_{\sigma,A}^-$ but its complement contains $P_{\sigma,A}^+$, then on the endpoint vector g_σ ,*

$$\frac{\|P_{\sigma,A}^+ \mathcal{T}_\sigma g_\sigma\|_2}{\|P_{\sigma,A}^- \mathcal{T}_\sigma g_\sigma\|_2} \longrightarrow 1. \quad (29)$$

Thus no estimate of the form

$$\|Q \mathcal{T}_\sigma P v\| \leq \varepsilon_\sigma \|P \mathcal{T}_\sigma P v\|, \quad \varepsilon_\sigma \rightarrow 0, \quad (30)$$

can hold for all packet vectors in the natural L^2 scaling. The same is true after multiplying an entire time slice by a Riccati balance weight.

The last sentence follows because the two branches occur in the same time slice and receive the same scalar time weight. One could suppress the sibling by assigning a singular branch-dependent norm, but the associated entrance/exit condition number would then have to be included explicitly.

Remark 3.3 (Far critical tails are different). The profile $\Gamma_{d,t}$ has quartic exponential decay: as $|q| \rightarrow \infty$, $A_d(q) \asymp q^2$, $B_d(q) \sim -A_d(q)$, and the Gaussian factor in (20) is e^{-cq^4} . Hence

$$\int_{|q-q_b|>A} \Gamma_{d,t}(q)^2 dq \longrightarrow 0 \quad (A \rightarrow \infty). \quad (31)$$

The complement can therefore be split into a controllable far critical tail and an uncontrollable-by-tail-estimates sibling channel. The theorem says that increasing the window does not remove the latter.

The sibling theorem concerns an off-diagonal coupling factor. By itself it does not prove that the full resolvent self-energy (3) is order one: re-entry and resolvent cancellation could still occur. The finite-horizon returns below test precisely that possibility numerically.

4 Exact branch splitting of a finite return

Let $P_0, \dots, P_{k-2}$ be the time-labeled packet-window projections and split the final window into disjoint pieces

$$P_{k-1}^{\text{both}} = P_{k-1}^- + P_{k-1}^+, \quad P_{k-1}^- P_{k-1}^+ = 0. \quad (32)$$

Here $-$ is the boundary-word branch $x < b$ and $+$ is its critical sibling. Define returns on $\text{ran } P_0$ by

$$R^\pm = P_0 \mathcal{T}_\sigma P_1 \mathcal{T}_\sigma \cdots P_{k-2} \mathcal{T}_\sigma P_{k-1}^\pm \mathcal{T}_\sigma P_0, \quad (33)$$

$$R^{\text{both}} = P_0 \mathcal{T}_\sigma P_1 \mathcal{T}_\sigma \cdots P_{k-2} \mathcal{T}_\sigma P_{k-1}^{\text{both}} \mathcal{T}_\sigma P_0, \quad (34)$$

$$R^{\text{full}} = P_0 \mathcal{T}_\sigma^k P_0. \quad (35)$$

Proposition 4.1 (Branch-sum identity and positivity order). *The branch-complete return satisfies the exact operator identity*

$$\boxed{R^{\text{both}} = R^- + R^+}. \quad (36)$$

All four returns are positive compact operators, and on nonnegative observables

$$0 \leq R^\pm \leq R^{\text{both}} \leq R^{\text{full}}. \quad (37)$$

Consequently their spectral radii obey

$$\text{spr}(R^\pm) \leq \text{spr}(R^{\text{both}}) \leq \text{spr}(R^{\text{full}}). \quad (38)$$

Proof. Insert (32) into (34) and use linearity to obtain (36). The Gaussian kernel is strictly positive and every window multiplier preserves the positive cone, so all returns are positive and compact. Removing a positive window restriction only adds nonnegative path contributions, which gives (37). Monotonicity of the spectral radius for positive compact operators proves (38) [1]. $\square$

Equation (36) does not imply $\text{spr}(R^{\text{both}}) = \text{spr}(R^-) + \text{spr}(R^+)$. The numerical near-additivity in the small-noise tail is evidence that the two positive returns have almost aligned principal endpoint vectors, consistent with their common Gaussian–quadratic origin.

5 The full time lift is a Floquet replication

The time labels used to make the local channels cyclic must not be confused with physical state variables. This distinction is exact at the level of the unrestricted operator.

Let T be a bounded operator on a complex Hilbert space $\mathcal{H}$. On $\mathcal{H}^k = \bigoplus_{j=0}^{k-1} \mathcal{H}$, define

$$(\mathfrak{C}_k v)_j = T v_{j+1 \pmod k}. \quad (39)$$

If U_k is the cyclic shift on $\mathbb{C}^k$, then

$$\mathfrak{C}_k = U_k \otimes T. \quad (40)$$

Write $\omega_k = e^{2\pi i/k}$.

Theorem 5.1 (Exact Floquet decomposition). *The discrete Fourier transform in the time coordinate gives the unitary equivalence*

$$\boxed{\mathfrak{C}_k \simeq \bigoplus_{\ell=0}^{k-1} \omega_k^\ell T}. \quad (41)$$

Therefore

$$\text{spec}(\mathfrak{C}_k) \setminus \{0\} = \bigcup_{\ell=0}^{k-1} \omega_k^\ell (\text{spec}(T) \setminus \{0\}). \quad (42)$$

If T is trace class, then

$$\boxed{\det_{\mathcal{H}^k}(\mathbf{I} - z\mathfrak{C}_k) = \prod_{\ell=0}^{k-1} \det_{\mathcal{H}}(\mathbf{I} - z\omega_k^\ell T) = \det_{\mathcal{H}}(\mathbf{I} - z^k T^k)}. \quad (43)$$

Proof. The Fourier vectors diagonalize U_k with eigenvalues ω_k^ℓ . Tensoring their decomposition with $\mathcal{H}$ proves (41) and (42). For a trace-class T , apply the Fredholm determinant to the direct sum and use

$$\prod_{\ell=0}^{k-1} (1 - z\omega_k^\ell \nu) = 1 - (z\nu)^k$$

for every eigenvalue ν of T , with algebraic multiplicity [2, 6]. $\square$

Take now $T = \mathcal{K}_\sigma^2$. If μ is one physical Markov eigenvalue, then $\nu = \mu^2$ is an eigenvalue of T , and the time lift contains

$$\omega_k^\ell \mu^2, \quad 0 \leq \ell < k. \quad (44)$$

The canonical bipartite lift

$$\mathfrak{B}_k = \begin{pmatrix} 0 & \mathbf{I} \\ \mathfrak{C}_k & 0 \end{pmatrix} \quad (45)$$

then contains

$$\pm \mu e^{\pi i \ell / k}, \quad 0 \leq \ell < k. \quad (46)$$

These $2k$ values arise from one physical μ .

Corollary 5.2 (Root-ring multiplicity no-go). *A theorem that identifies a $2k$ -point local root ring only with the spectrum of the sector-free bipartite time lift (45) cannot imply that $\mathcal{K}_\sigma$ has $2k$ distinct physical eigenvalues. Even a single simple physical eigenvalue generates such a ring by (46).*

This no-go statement is about inference, not about the existence of the observed cloud. The archived Markov computations do contain many distinct eigenvalues. The theorem says that time-lift root symmetry alone does not explain their multiplicity.

6 Exact Feshbach completion and its limitation

Return to the Gaussian operator and let

$$P = \text{diag}(P_0, \dots, P_{k-1}), \quad Q = \mathbf{I} - P \quad (47)$$

on $\mathcal{H}^k$. The local block from RH-18 is exactly

$$C_{PP} = P\mathfrak{C}_k P, \quad (C_{PP}v)_j = P_j \mathcal{T}_\sigma P_{j+1} v_{j+1}. \quad (48)$$

Write similarly $C_{PQ} = P\mathfrak{C}_k Q$, $C_{QP} = Q\mathfrak{C}_k P$, and $C_{QQ} = Q\mathfrak{C}_k Q$.

Theorem 6.1 (Time-lift Feshbach map). *Suppose $\zeta \notin \text{spec}(C_{QQ})$. Then $\zeta\mathbf{I} - \mathfrak{C}_k$ is invertible if and only if the effective operator on $\text{ran } P$,*

$$\mathcal{F}_P(\zeta) = \zeta P - C_{PP} - \Sigma_P(\zeta), \quad (49)$$

is invertible, where

$$\Sigma_P(\zeta) = C_{PQ}(\zeta Q - C_{QQ})^{-1} C_{QP}. \quad (50)$$

Algebraic multiplicities agree. In finite dimension, and for compatible Fredholm determinants in trace-class settings,

$$\det(\zeta\mathbf{I} - \mathfrak{C}_k) = \det(\zeta Q - C_{QQ}) \det \mathcal{F}_P(\zeta). \quad (51)$$

Proof. Write $\zeta\mathbf{I} - \mathfrak{C}_k$ in its $P \oplus Q$ block form and eliminate the Q variable. The Schur complement is (49). Standard block Gaussian elimination proves invertibility, multiplicity, and determinant statements [5, 7]. $\square$

The theorem is exact, but its target is the full time lift. Combining it with [theorem 5.1](#) gives the central limitation.

Theorem 6.2 (Sector-free Feshbach no-go). *Assume one proves, on contours around a local root ring, that $\Sigma_P(\zeta)$ is a sufficiently small perturbation of $\zeta P - C_{PP}$. The resulting eigenvalue count is a count for $\mathfrak{C}_k$, hence for the union of Floquet-rotated spectra in (42). Without an additional sector constraint, it does not determine how many distinct eigenvalues T has. After the bipartite lift, it likewise does not determine the physical resonance count of $\mathcal{K}_\sigma$.*

Proof. By [theorem 6.1](#), the completed effective determinant has exactly the zeros of $\mathfrak{C}_k$ not supplied by the complement factor. By [theorem 5.1](#), these zeros are Floquet rotations of the spectrum of T . The counterexample is already a T with one simple nonzero eigenvalue: its time lift has k rotated eigenvalues and its bipartite lift has the $2k$ values (46). Therefore the lift count does not determine the physical count. $\square$

Let Π_ℓ denote the orthogonal projection onto the ℓ th Fourier time sector. The physical information can be recovered from any fixed sector of the *full* lift, because $\Pi_\ell \mathfrak{C}_k \Pi_\ell \simeq \omega_k^\ell T$. The packet projection P , however, is phase dependent and generally

$$[P, \Pi_\ell] \neq 0. \quad (52)$$

Thus one cannot simply project the already localized cyclic block onto a sector. A corrected Grushin problem must carry the Floquet phase as a spectral parameter in its entrance, exit, or closing condition.

7 Peripheral deflation does not commute with localization

The physical bulk determinant is defined only after the Perron and parity modes are removed. Let E_0 and E_- be their spectral projectors for $\mathcal{K}_\sigma$, with eigenvalues 1 and $\lambda_-(\sigma)$, and define

$$K_b = \mathcal{K}_\sigma - E_0 - \lambda_- E_-. \quad (53)$$

The projectors commute with $\mathcal{K}_\sigma$, annihilate one another, and annihilate K_b . Hence

$$K_b^2 = \mathcal{K}_\sigma^2 - E_0 - \lambda_-^2 E_-. \quad (54)$$

Proposition 7.1 (Localized deflation identity). *For every pair of packet windows,*

$$\boxed{P_j K_b^2 P_{j+1} = P_j \mathcal{K}_\sigma^2 P_{j+1} - P_j E_0 P_{j+1} - \lambda_-^2 P_j E_- P_{j+1}.} \quad (55)$$

The difference has rank at most two, but it vanishes only under special orthogonality conditions. In particular, global spectral deflation cannot be moved through the packet projections unless the commutators $[P_j, E_0]$ and $[P_j, E_-]$ are controlled.

Proof. Multiply (54) on the left and right by the two window projections. The rank statement follows from the simplicity of the two peripheral modes. $\square$

Low rank is not the same as small norm. Endpoint windows are located exactly where the peripheral boundary layers are concentrated, so the correction in (55) can be leading. This is why a corrected Grushin construction must include peripheral extraction in its entrance and exit maps, rather than localize the positive kernel first and subtract two eigenvalues at the end.

8 Seven-noise audit

The analytic no-go results require no numerical hypothesis. The computation tests three narrower questions: whether the sibling remains comparable at the archived noise levels, whether it re-enters the same endpoint return, and how strongly peripheral deflation changes localized channels.

8.1 Protocol

We use the same row-normalized folded midpoint matrices as RH-18, with

$$n\sigma = 20.48 \quad (56)$$

and an eight-standard-deviation sparse Gaussian cutoff. The period is selected independently by $k = N_\sigma^H + 1$, where N_σ^H is the RH-16 Hellinger half-energy rank. All packet windows use the fixed multiple $L = 6$. The left return eigenvalues are imported unchanged from RH-18; right, two-branch, and unrestricted endpoint returns are recomputed by nine positive power iterations. The largest reported residual is 4.0×10^{-9} at the coarsest two-branch return and all tail residuals are near machine precision.

The sibling exit ratio is measured after one pullback of the truncated endpoint Gaussian. The “far” ratio is the mass outside the fixed two-branch window, not an asymptotic growing-window estimate. Its staircase variation reflects the changing value of δ_k/σ and should not be mistaken for a universal limit.

8.2 The sibling return is comparable

Table 1: One-step radii obtained from the principal eigenvalues of the left, right-sibling, branch-complete, and unrestricted endpoint returns. The last column is the archived full Markov bulk edge.

σ	k	left	right	both	full endpoint	bulk edge
10^{-2}	3	0.740767	0.650504	0.801218	0.819762	0.674669
4×10^{-3}	4	0.745592	0.700301	0.794838	0.808131	0.717941
2×10^{-3}	5	0.749691	0.747337	0.802324	0.802646	0.736278
10^{-3}	6	0.747069	0.747420	0.791673	0.791700	0.739142
5×10^{-4}	6	0.751346	0.707363	0.777588	0.794777	0.748015
2×10^{-4}	7	0.755108	0.735540	0.784378	0.792928	0.754749
10^{-4}	8	0.756459	0.756211	0.789825	0.789920	0.757023

At $\sigma = 10^{-4}$, the return eigenvalues before taking 16th roots are

$$\eta_- = 0.01149663526, \quad \eta_+ = 0.01143635812, \quad \eta_{\text{both}} = 0.02293473843, \quad \eta_{\text{full}} = 0.02297880610. \quad (57)$$

Thus

$$\frac{\eta_+}{\eta_-} = 0.99475698, \quad (58)$$

$$\frac{\eta_{\text{both}} - \eta_- - \eta_+}{\eta_{\text{both}}} = 7.61 \times 10^{-5}, \quad (59)$$

$$\frac{\eta_{\text{full}} - \eta_{\text{both}}}{\eta_{\text{full}}} = 1.92 \times 10^{-3}. \quad (60)$$

The operator sum (36) is exact; the small number in (59) says that the two principal return vectors are also almost aligned. Equation (60) says that, at this finite horizon, nearly the entire unrestricted positive endpoint return is already supplied by the two critical branches.

The single-pullback sibling/left norm ratios are

$$0.89295, 0.89606, 0.98581, 0.99030, 0.72342, 0.88317, 0.99738 \quad (61)$$

in decreasing noise order. The dips occur where the fixed six-width window cuts a larger part of one lobe; they do not contradict the symmetric-window theorem. At the clean tail endpoints the ratios approach one.

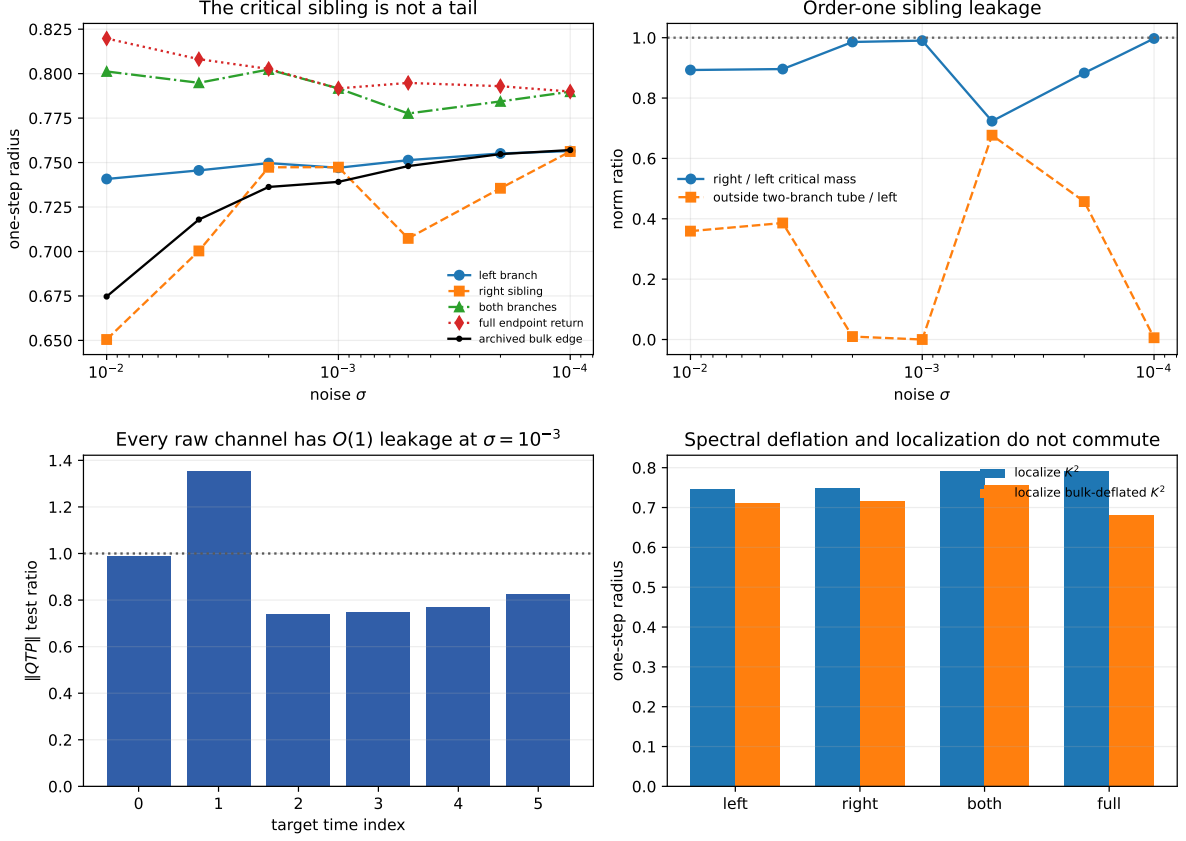


Figure 1: Complement-coupling audit. Top left: one-branch, two-branch, and unrestricted endpoint radii versus the archived physical bulk edge. Top right: sibling and fixed-window far exit masses relative to the retained branch. Bottom left: complement/local test ratios for all six component channels at $\sigma = 10^{-3}$. Bottom right: local returns formed before and after global Perron/parity deflation. The comparisons are floating-point diagnostics, not norm enclosures.

The branch-complete positive return lies near 0.79, above both the physical bulk edge and the limiting pole radius $\lambda^{-1/2} = 0.7718445\dots$. Therefore the successful left-branch radius cannot be justified by saying that all omitted positive paths are asymptotically negligible. Some sector-dependent cancellation or extraction must occur before the physical bulk determinant is obtained.

8.3 Deflation-localization benchmark

At $\sigma = 10^{-3}$, sparse right and left eigenvectors resolve

$$\lambda_0 = 1.0000000000000002, \quad \lambda_- = -0.996389276627418, \quad (62)$$

with all four left/right residuals below 2.1×10^{-15} . Applying the rank-two deflated operator before every localization gives the following.

Table 2: Noncommutation of global Perron/parity deflation and packet localization at $\sigma = 10^{-3}$, $k = 6$.

return	localize $\mathcal{K}_\sigma^2$	localize K_b^2
left	0.747069	0.712178
right	0.747420	0.716453
both	0.791673	0.756816
full endpoint	0.791700	0.681532

The shifts are far larger than the $1/k$ -scale radial precision sought from a perturbative cloud theorem. This one-noise benchmark is not an asymptotic claim; it demonstrates that the rank-two terms in (55) cannot simply be omitted from the proposed effective block.

8.4 Floquet and Schur algebra audit

A random real 3×3 operator at period five verifies the exact finite identities independently. The maximum eigenvalue matching error in (41) is 1.50×10^{-15} , the maximum relative determinant error in (43) is 1.11×10^{-16} , and the maximum Schur factorization error in (51) is 3.49×10^{-17} . For the deliberately non-invariant phase-dependent test projection, the self-energy/local norm ratio ranges from 1.57 to 4.30 at three nonsingular spectral parameters. These last random-matrix ratios are only a software test of a non-small regime, not evidence about the Gaussian asymptotic.

At $\sigma = 10^{-2}$, the leading physical Nyström eigenvalues are squared and rotated through the three component-time sectors. The right panel of figure 2 shows the exact replication. Each three-point ring comes from one point in the left panel.

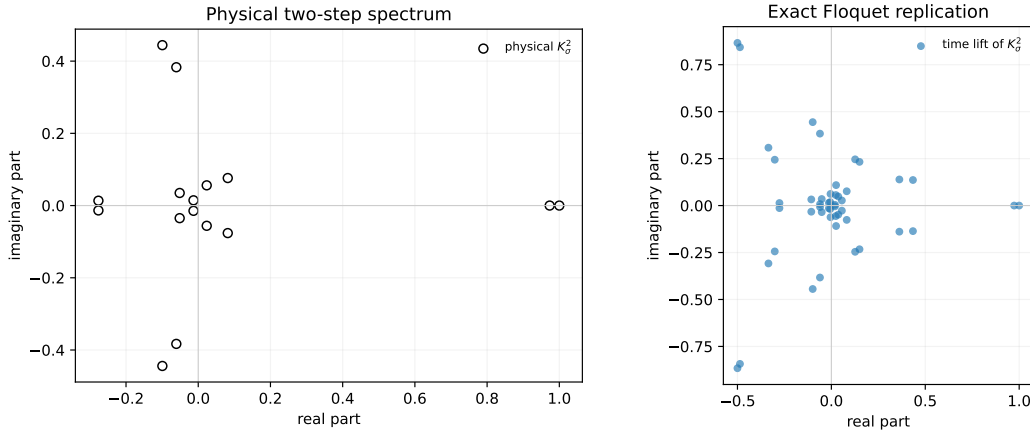


Figure 2: The physical two-step spectrum (left) and its exact time-unfolded Floquet replication at component period three (right). The extra rotated points are eigenvalues of the time lift, not additional physical Markov resonances.

8.5 Reproducibility and numerical status

The implementation tests the Fourier block decomposition, determinant identity, phase projections, exact Schur factorization, branch-sum identity, matrix-free endpoint returns, and rank-two deflation action. The full audit records all return values, residuals, leakage ratios, software versions, source hashes, peripheral residuals, Floquet copies, and figures. NumPy, SciPy, and Matplotlib provide the numerical implementation [3, 4, 8]. No calculation uses interval arithmetic or supplies a non-normal pseudospectral enclosure.

9 The corrected operator target

The failed strategy was specific:

$$\text{one branch} + \text{slowly growing windows} + \text{sector-free time lift} \implies? \text{ physical cloud block. (63)}$$

The critical-sibling theorem disproves small branch complement coupling, and the Floquet theorem disproves the final counting inference. A corrected construction must change all three ingredients.

- R1. Two-branch critical entrance.** Replace the scalar final packet by the vector $(\Gamma_{d,t}^-, \Gamma_{d,t}^+)$. The corresponding return object is at least 2×2 in branch space. Its symmetric and antisymmetric channels must be computed rather than selecting the left branch by hand.
- R2. Peripheral extraction inside the Grushin data.** Choose entrance and exit maps biorthogonal to E_0 and E_- before localization. Identity (55) then becomes part of the model, not a posteriori error.
- R3. Spectral-parameter closing phase.** Introduce a Floquet multiplier tied to the physical spectral parameter. The effective Hamiltonian should be a function $E_{-+}(\mu)$ whose closing condition selects a physical sector, rather than the spectrum of a fixed sector-free cyclic matrix.
- R4. Only then control the complement resolvent.** Decompose the remaining complement into far Gaussian tails and genuine bulk excursions. The former may admit growing-window estimates; the latter require a resolvent bound after the branch and peripheral channels have been removed.

A useful next theorem would derive the 2×2 critical return matrix and identify which branch combination carries the parity-extracted determinant. If one combination reproduces the RH-18 radius while the other is absorbed by peripheral extraction, the program can return to a sector-resolved Schur self-energy estimate. If not, the single-cycle spectral interpretation must be abandoned in favor of the already rigorous trace/determinant route.

10 Discussion

10.1 What the negative result changes

The RH-18 branch block remains a well-defined positive compact operator, and its principal radius remains an accurate numerical predictor of the outer bulk edge. What changes is the proposed proof mechanism. The omitted right branch has equal asymptotic critical mass and an almost equal finite return; therefore the left block is not isolated by Gaussian decay. Moreover, its exact root phases arise in a time-labeled space where such phases are present for every physical eigenvalue by Fourier replication.

Thus two visually successful facts—the left radial match and the exact local root ring—do not yet combine into a physical spectral theorem. This is precisely the kind of distinction a Feshbach analysis was meant to test. The result prevents a plausible but invalid proof from being built on top of the earlier numerical evidence.

10.2 What remains intact

The deterministic pole factor (1) is a trace/determinant theorem and is unaffected. The half-logarithmic endpoint rank is a singular-value theorem and is unaffected. The boundary-cycle multiplier, Riccati packet, conditioned critical profile, and local block root algebra are also

unchanged. The archived full Markov cloud consists of genuinely distinct computed eigenvalues; the Floquet theorem does not make those data artificial. It only shows that the time lift cannot certify their distinctness.

The numerical near-equality $R^{\text{full}} \approx R^- + R^+$ at the smallest noise is informative. It suggests that the first correction should be a finite branch matrix rather than an uncontrolled infinite complement. This is a narrower and more testable target than the original scalar Feshbach claim.

10.3 No arithmetic or self-adjoint implication

All operators here arise from one noisy quadratic map. The roots of unity in the time lift are a finite Floquet replication, not zeta zeros. No result constructs a self-adjoint Hilbert–Pólya operator, proves a prime-power trace formula, or implies the Riemann hypothesis. The no-go theorem is an operator-theoretic clarification internal to this dynamical model.

11 Conclusion

The complement gate gives a clear negative answer to the naive strategy. The critical sibling omitted by the branch-isolated return has the same asymptotic pullback mass as the retained branch, so off-block coupling cannot be reduced to a Gaussian tail. Independently, the full cyclic time lift is an exact direct sum of rotated physical two-step operators. Its root rings are Floquet copies and cannot certify physical resonance multiplicity.

The exact Feshbach map remains valid, but completing the scalar local block in the sector-free lift would solve the wrong counting problem. Global Perron/parity deflation also changes localized channels by leading rank-two terms. At the smallest computed noise, the right return is 99.48% of the left return and the two-branch return accounts for 99.81% of the unrestricted positive endpoint return, making the obstruction numerically visible as well as analytic.

The route is narrowed rather than closed. The next admissible object is a two-branch, peripherally extracted, Floquet-sector effective Hamiltonian. Its symmetric and antisymmetric branch channels must be resolved before any claim of small Schur self-energy or physical cloud multiplicity is made. This is the precise operator problem left by the present boundary test.

Data and code availability

The source code, tests, complete CSV and JSON audits, peripheral projectors, Floquet replication data, figures, and manuscript are available with this paper [11].

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